Special Relativity Notes

Covariant Derivative, Gradient, Divergence, Laplacian and Curl

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1 Covariant Derivative and Gradient

Write a vector in terms of the base vectors $\vec{e}_k$:

$$\vec{V} = V^k \, \vec{e}_k .$$

Differentiating the *vector* (components and base vectors) with respect to q^k gives the covariant derivative, whose components are written with a semicolon:

$$\frac{\partial \vec{V}}{\partial q^k} = V^j_{\;;k} \, \vec{e}_j .$$

Expanding with the product rule,

$$\frac{\partial \vec{V}}{\partial q^k} = \frac{\partial}{\partial q^k}\left(V^j \vec{e}_j\right) = \frac{\partial V^j}{\partial q^k} \, \vec{e}_j + V^j \, \frac{\partial \vec{e}_j}{\partial q^k} .$$

The derivative of a base vector is itself a vector, so it can be expanded in the same basis. The coefficients are the **Christoffel symbols**:

$$\frac{\partial \vec{e}_j}{\partial q^k} = \Gamma^{\mu}_{jk} \, \vec{e}_\mu .$$

Hence

$$\frac{\partial \vec{V}}{\partial q^k} = \frac{\partial V^j}{\partial q^k} \, \vec{e}_j + V^j \, \Gamma^{\mu}_{jk} \, \vec{e}_\mu .$$

The indices j and μ in the last term are dummy indices, so we may relabel $j \leftrightarrow \mu$:

$$\frac{\partial \vec{V}}{\partial q^k} = \frac{\partial V^j}{\partial q^k} \, \vec{e}_j + V^\mu \, \Gamma^{j}_{\mu k} \, \vec{e}_j = \left(\frac{\partial V^j}{\partial q^k} + \Gamma^{j}_{\mu k} \, V^\mu \right) \vec{e}_j = V^j_{\;;k} \, \vec{e}_j .$$

Therefore the **covariant derivative** of the components is

$$\boxed{V^j_{\;;k} = \frac{\partial V^j}{\partial q^k} + \Gamma^{j}_{\mu k} \, V^\mu}$$

which is *not* merely $\partial V^j / \partial q^k$.

Gradient. For a scalar ψ,

$$\vec{\nabla}\psi = \frac{\partial \psi}{\partial q^j}\,\vec{\epsilon}^{\,j}, \qquad \text{i.e.} \qquad \vec{\nabla} = \vec{\epsilon}^{\,j}\,\frac{\partial}{\partial q^j}.$$

Exercise. Γ^{μ}_{jk} is the Christoffel symbol, defined by $\partial\vec{\epsilon}_j/\partial q^k = \Gamma^{\mu}_{jk}\vec{\epsilon}_\mu$. Prove that it is **not** a tensor.

Proof. Go to new coordinates q', so $\vec{\epsilon}'_j = \dfrac{\partial q^a}{\partial q'^j}\vec{\epsilon}_a$. Then

$$\frac{\partial \vec{\epsilon}'_j}{\partial q'^k} = \frac{\partial^2 q^a}{\partial q'^k\,\partial q'^j}\,\vec{\epsilon}_a + \frac{\partial q^a}{\partial q'^j}\frac{\partial q^b}{\partial q'^k}\,\Gamma^{\mu}_{ab}\,\vec{\epsilon}_\mu.$$

Reading off the coefficient of $\vec{\epsilon}'_m$ (using $\vec{\epsilon}_\mu = \dfrac{\partial q'^m}{\partial q^\mu}\vec{\epsilon}'_m$),

$$\Gamma'^{\,m}_{jk} = \frac{\partial q'^m}{\partial q^\mu}\left[\frac{\partial^2 q^\mu}{\partial q'^j\,\partial q'^k} + \frac{\partial q^a}{\partial q'^j}\frac{\partial q^b}{\partial q'^k}\,\Gamma^{\mu}_{ab}\right].$$

The inhomogeneous second-derivative term spoils the tensor transformation law, so Γ^{μ}_{jk} is not a tensor. □

Exercise. Show that $V^{j}_{\;;k}$ transforms as a second-rank tensor.

Proof. $\partial\vec{V}/\partial q^k$ is the derivative of a vector (which is coordinate independent) with respect to a coordinate, so under $q \to q'$ it picks up exactly one factor $\partial q^b/\partial q'^k$. Its components $V^{j}_{\;;k} = \vec{\epsilon}^{\,j}\cdot\partial\vec{V}/\partial q^k$ pick up one factor $\partial q'^j/\partial q^a$ from $\vec{\epsilon}^{\,j}$. Thus

$$V'^{\,j}_{\;;k} = \frac{\partial q'^j}{\partial q^a}\,\frac{\partial q^b}{\partial q'^k}\,V^{a}_{\;;b},$$

the transformation law of a rank-2 (one contravariant, one covariant) tensor. The extra term in Γ cancels exactly the extra term in $\partial V^j/\partial q^k$. □

2 Christoffel Symbols of the First and Second Kind

Dot the defining relation $\partial\vec{\epsilon}_j/\partial q^k = \Gamma^{\mu}_{jk}\vec{\epsilon}_\mu$ with $\vec{\epsilon}^{\,m}$ and use $\vec{\epsilon}^{\,m}\cdot\vec{\epsilon}_\mu = \delta^m_\mu$:

$$\vec{\epsilon}^{\,m}\cdot\frac{\partial\vec{\epsilon}_j}{\partial q^k} = \Gamma^{\mu}_{jk}\,\delta^m_\mu \implies \boxed{\vec{\epsilon}^{\,m}\cdot\frac{\partial\vec{\epsilon}_j}{\partial q^k} = \Gamma^{m}_{jk}} \qquad \text{(second kind)}.$$

Lowering the index with g^{mn}: $g^{mn}\,\vec{\epsilon}_n\cdot\dfrac{\partial\vec{\epsilon}_j}{\partial q^k} = \vec{\epsilon}^{\,m}\cdot\dfrac{\partial\vec{\epsilon}_j}{\partial q^k} = \Gamma^{m}_{jk}$. Define

$$\boxed{[jk,n] = \vec{\epsilon}_n\cdot\frac{\partial\vec{\epsilon}_j}{\partial q^k}} \qquad \text{(first kind)}.$$

Exercise. How can the two kinds of Christoffel symbols be written in terms of each other?

$$g^{mn}\,[jk,n] = \Gamma^{m}_{jk}, \qquad\qquad [jk,n] = g_{mn}\,\Gamma^{m}_{jk}.$$

3 Christoffel Symbol in Terms of the Metric

Using $g_{ij} = \vec{\epsilon}_i \cdot \vec{\epsilon}_j$ (and $g^{ij} = \vec{\epsilon}^{\,i} \cdot \vec{\epsilon}^{\,j}$),

$$\frac{\partial g_{ij}}{\partial q^k} = \frac{\partial \vec{\epsilon}_i}{\partial q^k} \cdot \vec{\epsilon}_j + \vec{\epsilon}_i \cdot \frac{\partial \vec{\epsilon}_j}{\partial q^k} \implies \frac{\partial g_{ij}}{\partial q^k} = [ik, j] + [jk, i] \tag{1}$$

Relabelling the indices cyclically,

$$\frac{\partial g_{ik}}{\partial q^j} = [ij, k] + [kj, i], \tag{2}$$

$$\frac{\partial g_{kj}}{\partial q^i} = [ki, j] + [ji, k]. \tag{3}$$

Since $[jk, n] = \vec{\epsilon}_n \cdot \dfrac{\partial^2 \vec{r}}{\partial q^j \partial q^k}$ is symmetric in j, k, forming $(1) + (2) - (3)$ leaves $2[jk, i]$:

$$\boxed{[jk, i] = \frac{1}{2}\left[\frac{\partial g_{ij}}{\partial q^k} + \frac{\partial g_{ik}}{\partial q^j} - \frac{\partial g_{kj}}{\partial q^i}\right]}$$

This is the Christoffel symbol of the first kind in terms of derivatives of g_{ij}. The second kind follows from $\Gamma^m_{jk} = g^{mi}[jk, i]$.

Exercise. Do the same for g^{ij}.

4 Divergence of a Vector

With $\vec{V} = V^k \vec{\epsilon}_k$,

$$\vec{\nabla} \cdot \vec{V} = \vec{\epsilon}^{\,j} \cdot \frac{\partial \vec{V}}{\partial q^j} = \vec{\epsilon}^{\,j} \cdot V^k_{;j}\, \vec{\epsilon}_k = V^k_{;j}\, \delta^j_k = V^k_{;k} = \frac{\partial V^k}{\partial q^k} + \Gamma^k_{kj}\, V^j.$$

Step 1: the contracted Christoffel symbol

From $\Gamma^m_{jk} = g^{mi}[jk, i]$ and the first-kind formula,

$$\Gamma^k_{kj} = g^{km}\,[kj, m] = \frac{1}{2}\, g^{km}\left[\frac{\partial g_{mk}}{\partial q^j} + \frac{\partial g_{mj}}{\partial q^k} - \frac{\partial g_{kj}}{\partial q^m}\right].$$

The last two terms cancel: g^{km} is symmetric in (k, m), while $\partial_k g_{mj} - \partial_m g_{kj}$ is antisymmetric under $k \leftrightarrow m$. Hence

$$\Gamma^k_{kj} = \frac{1}{2}\, g^{km}\, \frac{\partial g_{km}}{\partial q^j}. \tag{4}$$

Step 2: derivative of $\det g$

Exercise. Show that $\dfrac{\partial}{\partial q^j}[\det g] = (\det g)\, g^{km}\, \dfrac{\partial g_{km}}{\partial q^j}$.

Proof. Expand $\det g$ along any row: $\det g = \sum_m g_{km} C^{km}$ with C^{km} the cofactors, which do not contain g_{km}. So $\partial(\det g)/\partial g_{km} = C^{km}$. For the inverse matrix, $g^{km} = C^{km}/\det g$ (the matrix is symmetric). Therefore, by the chain rule,

$$\frac{\partial(\det g)}{\partial q^j} = \sum_{k,m} \frac{\partial(\det g)}{\partial g_{km}}\, \frac{\partial g_{km}}{\partial q^j} = \sum_{k,m} (\det g)\, g^{km}\, \frac{\partial g_{km}}{\partial q^j}. \qquad \square$$

Combining with (4):

$$\Gamma^k_{kj} = \frac{1}{2}\,\frac{1}{\det g}\,\frac{\partial(\det g)}{\partial q^j} = \frac{1}{\sqrt{\det g}}\,\frac{\partial\sqrt{\det g}}{\partial q^j} = \frac{\partial}{\partial q^j}\ln\sqrt{\det g}.$$

Step 3: the divergence

$$\vec\nabla\cdot\vec V = \frac{\partial V^k}{\partial q^k} + V^j\,\frac{1}{\sqrt{\det g}}\,\frac{\partial\sqrt{\det g}}{\partial q^j} = \frac{1}{\sqrt{\det g}}\left[\sqrt{\det g}\,\frac{\partial V^k}{\partial q^k} + V^k\,\frac{\partial\sqrt{\det g}}{\partial q^k}\right]$$

(relabelling $j \to k$ in the second term), and the bracket is a product rule:

$$\boxed{\vec\nabla\cdot\vec V = \frac{1}{(\det g)^{1/2}}\,\frac{\partial}{\partial q^k}\left[(\det g)^{1/2}\,V^k\right]}$$

5 Laplacian

The Laplacian is the divergence of the gradient. The gradient $\vec\nabla\psi = \dfrac{\partial\psi}{\partial q^j}\vec\epsilon^{\,j}$ has contravariant components $(\nabla\psi)^k = g^{kj}\,\dfrac{\partial\psi}{\partial q^j}$, because $\vec\epsilon^{\,j} = g^{jk}\vec\epsilon_k$. Applying the divergence formula with $V^k = (\nabla\psi)^k$:

$$\boxed{\nabla^2\psi = \vec\nabla\cdot\vec\nabla\psi = \frac{1}{(\det g)^{1/2}}\,\frac{\partial}{\partial q^k}\left[(\det g)^{1/2}\,g^{kj}\,\frac{\partial\psi}{\partial q^j}\right]}$$

6 Curl of a Vector

In Cartesian coordinates

$$(\vec\nabla\times\vec V)_i = \begin{vmatrix}\hat\imath & \hat\jmath & \hat k\\ \partial_x & \partial_y & \partial_z\\ V_x & V_y & V_z\end{vmatrix} = \epsilon_{ijk}\,\partial_j V_k.$$

Step 1: why only derivatives of V_k appear

The covariant derivative of the *covariant* components is $V_{k;j} = \dfrac{\partial V_k}{\partial q^j} - \Gamma^m_{kj}\,V_m$. The curl needs the antisymmetric combination, and $\Gamma^m_{kj} = \Gamma^m_{jk}$ is symmetric (since $\partial\vec\epsilon_k/\partial q^j = \partial^2\vec r/\partial q^j\partial q^k$ is). So

$$V_{k;j} - V_{j;k} = \frac{\partial V_k}{\partial q^j} - \frac{\partial V_j}{\partial q^k},$$

and the Christoffel terms cancel.

Step 2: the general formula

In curvilinear coordinates the Levi-Civita *tensor* is $\epsilon^{ijk} = [ijk]/\sqrt{\det g}$, where $[ijk] = \pm1, 0$ is the permutation symbol (the factor $1/\sqrt{\det g}$ is what makes it transform as a tensor). Replacing $\partial_j V_k$ by $V_{k;j}$ in the Cartesian formula gives

$$\boxed{(\vec\nabla\times\vec V)^i = \frac{1}{\sqrt{\det g}}\,[ijk]\,\frac{\partial V_k}{\partial q^j}}$$

Note the derivative acts on the covariant components $V_k = g_{kj}V^j$; the notes' version with V^j should be read with lowered index.

7 Orthogonal Coordinates and Scale Factors

In orthogonal coordinates the line element is

$$d\vec{s} = h_1\,dq_1\,\hat{e}_1 + h_2\,dq_2\,\hat{e}_2 + h_3\,dq_3\,\hat{e}_3,$$

where $h_i = |\partial\vec{r}/\partial q_i|$ are the scale factors. Then $\vec{e}_i = h_i\hat{e}_i$, so

$$g_{ij} = h_i^2\,\delta_{ij}, \qquad g^{ij} = \frac{1}{h_i^2}\,\delta_{ij}, \qquad \sqrt{\det g} = h_1h_2h_3.$$

Writing physical components $\hat{V}_i = h_iV^i$ (so $V_i = h_i\hat{V}_i$ with lowered index), the general formulas reduce to

$$\begin{aligned}
\vec{\nabla}\psi &= \sum_i \frac{1}{h_i}\frac{\partial\psi}{\partial q_i}\,\hat{e}_i, \\
\vec{\nabla}\cdot\vec{V} &= \frac{1}{h_1h_2h_3}\left[\frac{\partial}{\partial q_1}(h_2h_3\hat{V}_1) + \frac{\partial}{\partial q_2}(h_1h_3\hat{V}_2) + \frac{\partial}{\partial q_3}(h_1h_2\hat{V}_3)\right], \\
\nabla^2\psi &= \frac{1}{h_1h_2h_3}\sum_i \frac{\partial}{\partial q_i}\left[\frac{h_1h_2h_3}{h_i^2}\,\frac{\partial\psi}{\partial q_i}\right], \\
(\vec{\nabla}\times\vec{V})_{\hat{1}} &= \frac{1}{h_2h_3}\left[\frac{\partial(h_3\hat{V}_3)}{\partial q_2} - \frac{\partial(h_2\hat{V}_2)}{\partial q_3}\right] \quad \text{(and cyclic)}.
\end{aligned}$$

For the curl, $(\vec{\nabla}\times\vec{V})^1 = \dfrac{1}{h_1h_2h_3}\big(\partial_2V_3 - \partial_3V_2\big)$ and the physical component is $h_1(\vec{\nabla}\times\vec{V})^1$.

Example (spherical coordinates). $(q_1,q_2,q_3) = (r,\theta,\phi)$ with $h_1 = 1,\ h_2 = r,\ h_3 = r\sin\theta$ gives

$$\nabla^2\psi = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\psi}{\partial\phi^2}.$$